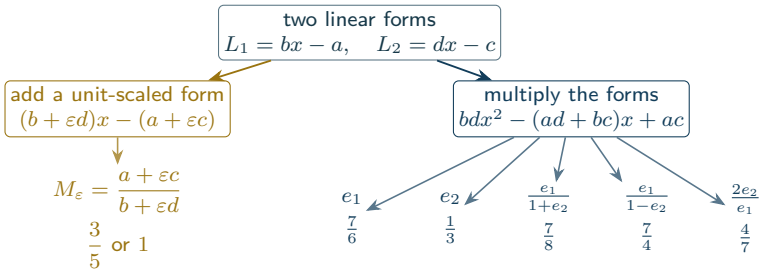
ARITHMETIC ON ONE PAGE

The split on other rings and other operations

Vieta fixes the sum and product; changing the allowed units changes the mediant.

1 / THE RULE AND THE PICTURE

Two operations on forms, many operations on fractions



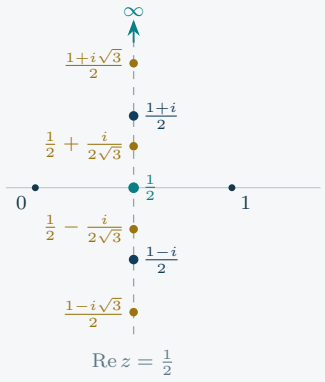
The roots are $\alpha = a/b$, $\beta = c/d$, with $b, d \neq 0$. Vieta reads off their sum and product:

$$e_1 = \frac{ad + bc}{bd}, \qquad e_2 = \frac{ac}{bd}.$$

These depend only on the values. The mediant depends on the chosen representatives.
The leaves use $\alpha = \frac{1}{2}$, $\beta = \frac{2}{3}$.

2 / ANOTHER RING: ONE MEDIANT FOR EACH UNIT

Changing representatives changes the mediant



A **unit** has an inverse in its ring. Reduced pairs rescale only by units:

$$M_\varepsilon(a/b, c/d) = \frac{a + \varepsilon c}{b + \varepsilon d}.$$

Distinct inputs give one different result per unit, including ∞ for zero denominator. For $\frac{1}{2}, \frac{2}{3}$, the units ± 1 give $\frac{3}{5}$ and 1.

ring	units ε	count
$\mathbb{Z}$	± 1	2
$\mathbb{Z}[i]$	$\pm 1, \pm i$	4
$\mathbb{Z}[\omega]$	$\pm 1, \pm\omega, \pm\omega^2$	6
$\mathbb{F}_7$	$1, 2, 3, 4, 5, 6$	6
$\mathbb{Z}[\varphi]$	$\pm\varphi^n,\ n \in \mathbb{Z}$	∞

Here $\omega = e^{2\pi i/3}$, $\varphi = (1 + \sqrt{5})/2$. Replacing (c, d) by $(a + \varepsilon c, b + \varepsilon d)$ changes $ad - bc$ to $\varepsilon(ad - bc)$; unit-determinant neighbours stay neighbours.
For $0, 1$, $M_\varepsilon = \varepsilon/(1 + \varepsilon)$ has real part $\frac{1}{2}$ when $|\varepsilon| = 1$, $\varepsilon \neq -1$. Colours: integers (teal), Gaussian (navy), Eisenstein (gold).

3 / ANOTHER OPERATION: SYMMETRIC FUNCTIONS OF THE ROOTS

The quadratic supplies a family of value-based operations

Every symmetric rational function of two roots can be written using $e_1 = \alpha + \beta$ and $e_2 = \alpha\beta$. For $\frac{1}{2}, \frac{2}{3}$, the quadratic is $6x^2 - 7x + 2$: $e_1 = \frac{7}{6}$ and $e_2 = \frac{1}{3}$.

operation over $\mathbb{Q}$	in e_1, e_2	at $\frac{1}{2}, \frac{2}{3}$	associative?
sum	e_1	$\frac{7}{6}$	yes
product	e_2	$\frac{1}{3}$	yes
Einstein sum	$\frac{e_1}{1+e_2}$	$\frac{7}{8}$	yes, where defined
tangent sum	$\frac{e_1}{1-e_2}$	$\frac{7}{4}$	yes, where defined
arithmetic mean	$\frac{1}{2}e_1$	$\frac{7}{12}$	no
harmonic mean	$\frac{2e_2}{e_1}$	$\frac{4}{7}$	no

The family $(u + v)/(1 - \kappa uv)$ gives tangent, ordinary and Einstein addition for $\kappa = 1, 0, -1$; Einstein inputs are normalised speeds.

4 / A FINITE FIELD, AND WHAT CARRIES OVER

Six mediants visit all the other points of $\mathbb{P}^1(\mathbb{F}_7)$

Modulo 7, $\frac{1}{2} = 4$, $\frac{2}{3} = 3$, $e_1 = 0$, $e_2 = 5$. Use $M_\varepsilon = (1 + 2\varepsilon)/(2 + 3\varepsilon)$:

ε	1	2	3	4	5	6
M_ε	2	5	0	∞	6	1

These are exactly the points other than the inputs $3, 4$. Division requires a nonzero denominator in a field, or a unit in a ring; a mean involving $1/2$ also requires 2 to be invertible.

ASIDE / TRY THESE

For $\frac{1}{3}, \frac{1}{2}$, find the two mediants $\frac{2}{5}, 0$. Check the Einstein result $\frac{7}{8}$ for $\frac{1}{2}, \frac{2}{3}$.

LINKS TO OTHER SHEETS

Velocities and means supply new leaves. Gaussian and Eisenstein units are polygon roots; φ has infinite order.

Conventions. Use distinct inputs and reduced representatives; finite-root formulas require $b, d \neq 0$. Real betweenness for the ordinary mediant requires denominators of the same sign. Polynomial expansion works over any commutative ring.

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